MILITARY SPECIFICATION

PROJECTILE, 155MM, CS, TACTICAL, XM631 ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers assembling and packing for one type of projectile designated as Projectile, 155MM, CS, Tactical, XM631.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1168 — Lot Numbering of Ammunition.

MIL-STD-1235 — Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

7549275 — Pallet for 155MM Projectile.

9220382 — Projectile, 155MM, CS, Tactical, XM631, Loading Assembly.

PUBLICATIONS

ARMY

ET-9220382 - Equipment Tabulation.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** Material and parts shall be in accordance with applicable drawings and specifications.

FSC 1320

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3.2 Projectile. The projectile shall comply with all requirements specified on Drawing (dwg.) 9220382 and with all requirements specified in applicable specifications.

3.3 Proving ground. The projectile shall comply with the following requirements when tested as specified in 4.3.1.

3.3.1 *Functioning.* The projectile shall function satisfactorily (see 6.5) and shall burn for the minimum time of 50 seconds.

3.3.2 *Premature bursts and security of parts.* There shall be no premature burst or indication of metal parts failure in the gun bore or in the first 100 yards of horizontal range.

3.4 Workmanship. All manufacturing, processing, and assembly operations of the projectile shall be correctly performed in a thoroughly workmanlike manner. Parts shall be free of burrs, chips, dirt, cracks, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define the terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity

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of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* The term "lot" as used throughout this specification refers to an inspection lot which is defined as an essentially homogeneous collection of units of products from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL–STD–105 and shall be numbered in accordance with MIL–STD–1168. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Each lot shall contain:

(a) Metal parts from one lot interfix number from one manufacturer.

(b) Ejection charges of one lot interfix number from one manufacturer.

(c) Canisters loaded at one plant using the same formula and process under the same drawing and revision and the same specification and revision.

(d) Projectile of one weight zone.

4.2.2 *Examination.* Inspection for Critical defects (and Major defects when so specified) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL–STD–105 except that continuous sampling plans in accordance with MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Projectile, prior to assembly of expelling charge, (see dwg. 9220382).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined.			
Major:	AQL 0.40 percent		
	102. Foreign matter in cavity	Visual	01001
Minor: None defined.			

4.2.2.2 Projectile, prior to assembly of base plug (see dwg. 9220382).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined.			
Major:	AQL 0.40 percent		
	101. Any component missing, improperly assembled or damaged	Visual	02001
Minor:	AQL 0.65 percent		
	201. Evidence of poor workmanship (see 3.4)	Visual	02002

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4.2.2.3 Projectile, prior to assembly of lifting plug (see dwg. 9220382).

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Projectile zone weight incorrect	Balance	03001
2.	Zone stenciling incorrect or punch marks missing or inadequate	Visual	03002
3.	Shear pin missing	Visual	03003
Major:	AQL 0.40 percent		
101.	Gap between base plug and body excessive	Gage	03004
Minor:	AQL 1.5 percent		
201.	Shear pin above flush	Visual	03005
202.	Shear pin loose or stake missing	Visual-Manual	03006
203.	Evidence of poor workmanship (see 3.4)	Visual	03007

4.2.2.4 Projectile loading assembly (see dwg. 9220382.

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Torque test failure of lifting plug	Test	04001
102.	Diameter of bourrelet, maximum	Gage	04002
103.	Lifting plug gasket missing	Visual	04003
104.	Grommet missing, damaged or improperly assembled	Visual	04004
Minor:	AQL 1.5 percent		
201.	Protective coating with total damaged area in excess of ¼ square inches	Visual	04005
202.	Marking missing, misleading or unidentifiable (except for zones)	Visual	04006
203.	Evidence of poor workmanship (see 3.4)	Visual	04007

4.2.2.5 *Pallet* (see dwg. 7549275).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Pallet damaged	Visual	05001
102.	DOD symbol missing or unidentifiable	Visual	05002
103.	Strapping missing or loose	Manual-Visual	05003
104.	Projectile of different zone on same pallet	Visual	05004
Minor:	AQL 1.5 percent		
201.	Number of projectile assemblies incorrect	Visual	05005
202.	Marking missing, misleading or unidentifiable	Visual	05006
203.	Strapping improperly engaged	Visual	05007

4.2.3 *Testing*

4.2.3.1 *Proving ground.* The following test shall be performed as specified in 4.3.1.

4.2.3.1.1 *Functioning (see 3.3.1) — Major defect — Code No. 04009.* Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirement, 27 projectiles shall be selected from each lot for this test. Twenty-five of the projectiles shall be fired for determination of satisfactory functioning (ejection, ignition and burning time of the 125 canister assemblies). If a total of 11 or more canister assemblies fail to eject, ignite or burn for the required time, the lot shall be rejected. (In the event that one or two projectiles fail to show evidence of fuze functioning, additional projectile(s) shall be fired in order that a total of 125 canister assemblies be observed for ejection, ignition, and burning time.)

4.2.3.1.2 After the three consecutive lots have met the criteria of 4.2.3.1.1, 11 projectiles shall be selected from each lot for this test. Ten of the projectiles shall be fired for determination of satisfactory functioning (ejection, ignition and burning time of the 50 canister assemblies). If a total of 6 or more canister assemblies fail to eject, ignite or burn for the required time, the lot shall be rejected. (In the event that one of the projectiles fails to show evidence of fuze functioning, additional projectile(s) shall be fired in order that a total of 50 canister as-

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semblies be observed for ejection, ignition, and burning time.)

4.2.3.1.3 *Premature burst (see 3.3.2) — Critical defect — Code No. 04010.* If any of the sample projectiles used in any of the tests fail to comply with the requirements for premature burst, the lot shall be rejected.

4.2.3.1.4 *Refiring.* If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional rounds as required.

4.2.4 *Inspection Equipment.* Equipment Tabulation Number ET–9220382 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection equipment.* The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 *Proving ground.* These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample projectiles shall be assembled with live fuzes and fired with M4A1 Propelling Charges at zone 7 in a weapon for which the ammunition is standard (preferably the M126 Howitzer) and be observed for conformance to all requirements. When samples are selected in accordance with 4.2.3.1.1, five shall be fired at ambient temperature, ten shall be conditioned at 145 ± 2 degrees Fahrenheit for 18 hours, minimum, and ten shall be conditioned at −40 ± 2 degrees Fahrenheit for 18 hours, minimum, prior to firing with testing to be conducted as soon as possible after removal of the samples from conditioning chambers. Sample items selected in accordance with 4.2.3.1.2 shall be tested at ambient temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 *Level A.* The projectiles shall be packed and marked in accordance with dwg. 7549275.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL–STD–1167.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections

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herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA–1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD — Contractor design on controlled contractor format and/or commercial equipment.

MU — Army design, mandatory for use.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and nonmandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL–D–1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 *Submission of contractor designs.* Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND1. All other designs of inspection equip-

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ment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of test data. In addition to normal distribution of records, when the projectile is procured by the Department of the Army, one copy of all ballistic data will be forwarded to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2, Dover, New Jersey, 07801.

6.5 Functioning. Satisfactory functioning shall be defined as functioning of the fuze, and ejection, ignition, and burning of the canister assembly for the required time without detonation. (Detonation is disintegration of the canister assembly in the air after ejection from the projectile which results in a puff of smoke or dust cloud.)

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1320-A119

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